

REMOTE-CONTROLLED SMARTFAN Using AT89C2051

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The remote-controlled smartfans available in the market are very expensive and their operating principles are difficult to understand. Here is a simple circuit for a remote-controlled smartfan that can control the speed of a fan or light intensity of an electric bulb.

The VIRE remote (Fig. 1) is generally used for MP3 players, FM radios and the like. Its range is up to 12 metres, so there is no need to control the fan/bulb manually. The AT89C2051 microcontroller (MCU) acts as a smartregulator.

Circuit and working

Circuit of the remote-controlled smartfan using AT89C2051 is shown in Fig. 2. It is built around centre-tapped 12V-0-12V, 750mA transformer (X1), rectifier diodes 1N4007 (D1, D2), 2200µF/25V (C1), 100µF/16V(C2) filter capacitors and voltage regulator 7805 (IC1) that produces 5V constant output voltage, AT89C2051 (IC2) MCU, IR receiver TSOP1738 (IRX1), VIRE IR remote, 12V electromagnetic relays, transistors BC547 (T1, T2, T3) and a few other components.

In this project, MCU acts as a smartelectronic regulator based on IR remote signals. It is a 20-pin MCS-51 family controller and has 2kB flash ROM, 128 bytes RAM, UART



Fig. 1: Screenshot of VIRE IR remote

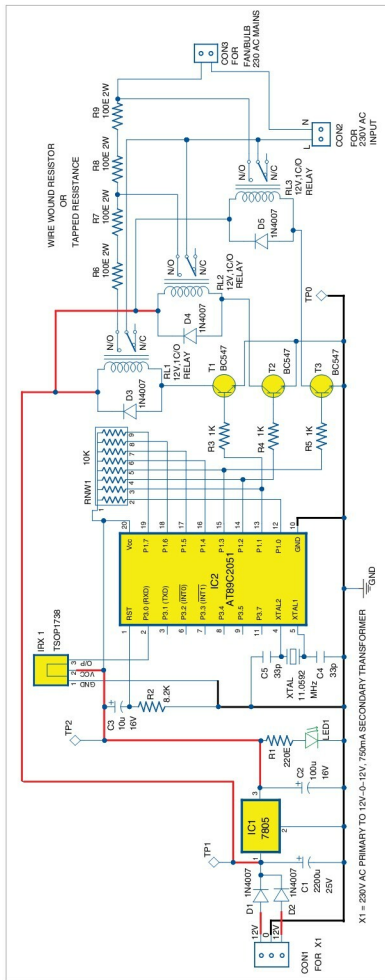


Fig. 2: Circuit of the remote-controlled smartfan using AT89C2051